

1. Introduction

1.1 Purpose

This test plan details the testing strategy for the different features and requirements for SplitsFromVideo. The purpose of this is to make sure that all the requirements outlined in the requirements document and design document are properly tested to ensure functionality and allow for proper implementation and cohesion.

1.2 Scope & Items to be Tested

Items Included: YouTube video processing (yt-dlp), UI rendering (Pywebview), interactive bounding box selection, OpenCV frame processing, PaddleOCR digit extraction, and file generation (.csv, .lss, .txt).

Items Excluded: External YouTube server maintenance and hardware-level GPU acceleration configuration.

1.3 Pass/Fail Criteria

Pass: The test case executes to completion, outputs expected results matching SRS criteria, and generates no unhandled exceptions.

Fail: System crashes, unhandled exceptions occur, output data is missing/corrupted, or performance metrics fall below SRS thresholds.

2. Test Approach & Environment

Testing Levels: Unit Testing (Python functions), Integration Testing (Pywebview <-> OpenCV <-> PaddleOCR pipeline), System Testing (End-to-End user flow).

Environment:

OS: Windows 10/11, macOS 10.15 or higher, Ubuntu 22.04 LTS.

Network: Broadband connection (10 Mbps minimum) and throttled high-latency/offline test environments.

3. Test Case Specifications

Test ID	Requirement(s)	Test Scenario	Input Data	Test Type	Expected Result
SI-02, FR-01, FR-02	CI-01, CI-02	Inputting and viewing YouTube video	Standard YouTube link	Usual	Video stream from YouTube containing requested video
FR-03, FR-04	SI-01, SI-02	Selecting timer region	Standard YouTube link, Area of timer	Usual	Video stream initializes, video player shows initial frame, allows for selecting box region within video
FR-05 through FR-12	SI-03, SI-04	Extracting split times from given YouTube video	Standard YouTube link, Area of timer, End of run	Usual	Downloadable file in selected format with extracted split times